

Week 12 Workshop Solutions

Question 1

Let $\omega_m = \$1m$ invested in Motorola, $\omega_c = \$1m$ invested in Citigroup.

a. The 1-step ahead forecast is

$$\begin{aligned}\hat{\sigma}_{m,t+1}^2 &= (1 - 0.96) r_{m,t}^2 + 0.96 \hat{\sigma}_{m,t}^2 \\ &= (1 - 0.96) 0.0629^2 + 0.96 (0.00303) \\ &= 0.00307\end{aligned}$$

The 5% VaR is

$$\begin{aligned}VaR_{0.05}^m &= |\omega_m (-1.645) \hat{\sigma}_{m,t+1}| \\ &= \left| 1000000 (-1.645) \sqrt{0.00307} \right| \\ &= \$91145.47026594356\end{aligned}$$

b. The 1-step ahead forecast is

$$\begin{aligned}\hat{\sigma}_{c,t+1}^2 &= (1 - 0.94) r_{c,t}^2 + 0.94 \hat{\sigma}_{c,t}^2 \\ &= (1 - 0.94) (-0.0133)^2 + 0.94 (0.0103) \\ &= 0.00969\end{aligned}$$

The 5% VaR is

$$\begin{aligned}VaR_{0.05}^c &= |\omega_c (-1.645) \hat{\sigma}_{c,t+1}| \\ &= \left| 1000000 (-1.645) \sqrt{0.00969} \right| \\ &= \$161930.17708259323\end{aligned}$$

c. Under the RiskMetrics

$$r_{pt}|F_{t-1} \sim N(0, \sigma_{p,t}^2)$$

where

$$\sigma_{p,t}^2 = Var(r_{pt}|F_{t-1}) = \omega^2 \sigma_{m,t}^2 + (1 - \omega)^2 \sigma_{c,t}^2 + 2\omega(1 - \omega) \rho \sigma_{m,t} \sigma_{c,t}$$

where $\rho = 0.411$. The 5% VaR is thus

$$VaR_{0.05} = |(\omega_m + \omega_c) \sigma_{p,t+1} (-1.645)|$$

The 1-step ahead forecast is

$$\begin{aligned}\hat{\sigma}_{p,t+1}^2 &= \omega^2 \hat{\sigma}_{m,t+1}^2 + (1 - \omega)^2 \hat{\sigma}_{c,t+1}^2 + 2\omega(1 - \omega) \hat{\rho} \hat{\sigma}_{m,t+1} \hat{\sigma}_{c,t+1} \\ &= 0.5^2 (0.00307) + 0.5^2 (0.00969) + 2(0.5)(0.5)(0.411) \sqrt{0.00307} \sqrt{0.00969} \\ &= 0.004310838\end{aligned}$$

The 5% VaR is thus

$$\begin{aligned}VaR_{0.05} &= |(\omega_m + \omega_c) (-1.645) \sigma_{p,t+1}| \\ &= \left| (1000000 + 1000000) (-1.645) \sqrt{0.004310838144459315} \right| \\ &= \$216011.4421956\end{aligned}$$

d. Using the formula

$$\begin{aligned} & \sqrt{91145.47026594356^2 + 161930.17708259323^2 + 2(0.411)(91145.47026594356)(161930.17708259323)} \\ &= 216011.4421956 \end{aligned}$$

e. Notice that

$$\omega = \frac{\omega_m}{\omega_m + \omega_c}$$

and so

$$\omega_m^2 = \omega^2 (\omega_m + \omega_c)^2$$

Also

$$1 - \omega = \frac{\omega_c}{\omega_m + \omega_c}$$

and so

$$\omega_c^2 = (1 - \omega)^2 (\omega_m + \omega_c)^2$$

The square of 5% VaR for the portfolio is

$$\begin{aligned} (VaR_{0.05})^2 &= (\omega_m + \omega_c)^2 \sigma_{p,t+1}^2 (-1.645)^2 \\ &= (\omega_m + \omega_c)^2 (-1.645)^2 \left(\omega^2 \sigma_{m,t+1}^2 + (1 - \omega)^2 \sigma_{c,t+1}^2 + 2\omega(1 - \omega) \rho \sigma_{m,t+1} \sigma_{c,t+1} \right) \\ &= (\omega_m + \omega_c)^2 \omega^2 (-1.645)^2 \sigma_{m,t+1}^2 \\ &\quad + (\omega_m + \omega_c)^2 (1 - \omega)^2 (-1.645)^2 \sigma_{c,t+1}^2 \\ &\quad + (\omega_m + \omega_c)^2 (-1.645)^2 2\omega(1 - \omega) \rho \sigma_{m,t+1} \sigma_{c,t+1} \\ &= \omega_m^2 (-1.645)^2 \sigma_{m,t+1}^2 \\ &\quad + \omega_c^2 (-1.645)^2 \sigma_{c,t+1}^2 \\ &\quad + 2\rho(\omega_m + \omega_c)\omega(-1.645)\sigma_{m,t+1}(\omega_m + \omega_c)(1 - \omega)(-1.645)\sigma_{c,t+1} \\ &= \omega_m^2 (-1.645)^2 \sigma_{m,t+1}^2 \\ &\quad + \omega_c^2 (-1.645)^2 \sigma_{c,t+1}^2 \\ &\quad + 2\rho|\omega_m(-1.645)\sigma_{m,t+1}||\omega_c(-1.645)\sigma_{c,t+1}| \\ &= (\text{VaR}_{0.05}^m)^2 + (\text{VaR}_{0.05}^c)^2 + 2\rho\text{VaR}_{0.05}^m\text{VaR}_{0.05}^c \end{aligned}$$

Question 2

The fitted GJR-GARCH model:

$$\begin{aligned} \hat{\sigma}_t^2 &= \hat{\alpha}_0 + \hat{\alpha}_1 \hat{\varepsilon}_{t-1}^2 + \hat{\lambda} (I_{t-1} \hat{\varepsilon}_{t-1}^2) + \hat{\beta}_1 \hat{\sigma}_{t-1}^2 \\ &= 0.024 + 0.0317 + 0.057 (I_{t-1} \hat{\varepsilon}_{t-1}^2) + 0.933 \hat{\sigma}_{t-1}^2 \end{aligned}$$

where $\hat{\varepsilon}_t = r_t - \hat{c} = r_t - 0.051$ and $I_{t-1} = 1$ when $\hat{\varepsilon}_{t-1} \leq 0$ and $I_{t-1} = 0$ when $\hat{\varepsilon}_{t-1} > 0$.

Since $\hat{\lambda} > 0$ this indicates the presence of a leverage effect: bad news ($\hat{\varepsilon}_{t-1} < 0$) has a greater impact on volatility $\hat{\sigma}_t^2$ than good news ($\hat{\varepsilon}_{t-1} > 0$), because $\hat{\alpha}_1 + \hat{\lambda} > \hat{\alpha}_1$.

The leverage effect is statistically significant, as the t-statistic for testing $H_0 : \lambda = 0$ vs $H_A : \lambda \neq 0$ is 4.06 which exceeds the critical value of 1.96. The p-value is 0, which is less than the 0.05 significance level.